SOC Detection Posture Assessment Report

1. Objective

The objective of this assessment is to evaluate the organization's current **detection posture** across email, endpoint, and identity domains. The focus is strictly on determining whether claimed detection capabilities are supported by observable telemetry and enabled detection logic, and whether such detections are available, controllable, and actionable at the SOC level.

This assessment does **not** evaluate analyst performance, incident response execution, or tool efficacy in isolation. The intent is to identify architectural strengths, constraints, and gaps in detection visibility and ownership.

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2. Assessment Perspective

This assessment evaluates detection capability from the perspective of **SOC visibility and controllability**. While upstream security controls may generate detections or take enforcement actions, detections that cannot be independently evaluated, contextualized, or extended at the SOC layer are treated as architectural constraints rather than complete detection coverage.

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3. Scope

3.1 In Scope

- Detection of malicious or suspicious email delivery
- Detection of anomalous or abusive authentication activity
- Detection of suspicious or malicious endpoint activity

3.2 Out of Scope

- Incident response execution and containment
- Malware reverse engineering
- Explicit threat hunting activities
- Detection tuning, automation, or SOAR workflows
- Alert volume, false-positive rates, or analyst workload

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4. Assumptions

- The assessment assumes existing SOC operating procedures are followed.
- Absence of relevant telemetry indicates lack of **potential** detection capability for the associated behavior.
- Presence of telemetry without enabled analytics indicates a detection gap.
- The assessment focuses on initial access and early-stage attacker activity; later-stage activities are considered only where directly enabled by identified gaps.

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5. Detection Status Definitions

Status	Definition
Detected	Enabled analytics generate alerts for the behavior without manual intervention
Visible but Undetected	Telemetry exists but no analytics convert it into alerts
Blind Spot	Required telemetry is absent

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6. Detection Coverage Evaluation

6.1 Claimed Detection Coverage

Domain	Claimed Behavior
Email	Detection of malicious email delivery including phishing, malicious URLs, and attachments
Endpoint	Detection of suspicious or malicious endpoint activity including execution and communication abuse
Identity	Detection of anomalous authentication activity and abnormal sign-in patterns

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6.2 Telemetry Validation

Domain	Telemetry Present	Notes
Email	Yes	Message, URL, and attachment telemetry ingested and continuous
Endpoint	Yes	Process, network, and device activity

Domain	Telemetry Present	Notes
		telemetry ingested
Identity	Yes	Authentication and sign-in telemetry ingested and continuous

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6.3 Detection Validation

Domain	Telemetry Exists	Analytics Logic Exists	Detection Status	Detection Logic Location
Email	Yes	No	Visible but Undetected	None
Endpoint	Yes	Yes	Detected	Upstream security control
Identity	Yes	Yes	Detected	SIEM-native

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7. Part 3 — Detection Gaps and Attack Advantage

7.1 Gap Summary

Gap ID	Domain	Gap Type	Detection Status	Attacker Advantage
GAP-EMAIL-01	Email	Detection Gap	Visible but Undetected	Initial access via phishing without SOC awareness
GAP-ENDPOINT-01	Endpoint	Control Ownership Gap	Detected	Limited visibility into pre-verdict endpoint activity
GAP-IDENTITY-01	Identity	Explainability Gap	Detected	Inconsistent triage of subtle identity abuse

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7.2 Detailed Gap Descriptions

GAP-EMAIL-01

Gap Description: Malicious or novel phishing emails can be delivered without SIEM-detectable alerts due to the absence of email delivery analytics, with detection dependent on upstream verdicts and user reporting.

Gap Type: Detection Gap
Attack Advantage: Enables initial access without immediate SOC visibility, increasing dwell time and likelihood of credential compromise.

GAP-ENDPOINT-01

Gap Description: Suspicious endpoint behavior preceding malware or exploit classification is not independently observable or evaluable within the SIEM due to reliance on upstream verdict-based alerts.

Gap Type: Control Ownership Gap
Attack Advantage: Allows reconnaissance or staging activity to occur with limited SOC context prior to verdict-based detection.

GAP-IDENTITY-01

Gap Description: A portion of identity anomaly detection relies on probabilistic analytics with limited behavioral explainability.

Gap Type: Explainability Gap
Attack Advantage: Reduces analyst confidence and consistency when handling subtle identity abuse scenarios.

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8. Part 4 — Risk Prioritization

Gap ID	Attack Surface	Detection Failure Mode	Blast Radius	Relative Risk
GAP-EMAIL-01	External	No Detection	Multiple Users	High
GAP-ENDPOINT-01	Internal	Delayed Context	Organization-wide	Medium

Gap ID	Attack Surface	Detection Failure Mode	Blast Radius	Relative Risk
GAP-IDENTITY-01	External	Low-Confidence Detection	Organization-wide	Medium

Risk Summary: Email-based initial access represents the highest relative risk due to lack of SOC-detectable delivery analytics. Endpoint and identity risks are mitigated by existing detections but remain material due to architectural constraints affecting visibility and explainability.

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9. Recommendations and Roadmap

9.1 Recommendation Mapping

Recommendation ID	Gap ID	Summary	Priority
REC-EMAIL-01	GAP-EMAIL-01	Introduce SIEM-side analytics for malicious email delivery	High
REC-ENDPOINT-01	GAP-ENDPOINT-01	Correlate endpoint verdicts with behavioral context	Medium
REC-IDENTITY-01	GAP-IDENTITY-01	Improve explainability of identity anomaly detections	Medium

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9.2 Detailed Recommendations

REC-EMAIL-01

Objective: Reduce reliance on upstream verdicts and user reporting for email-based initial access detection.
Action: Implement SIEM-side analytics evaluating email delivery telemetry to identify suspicious or malicious messages.
Expected Outcome: Independent SOC visibility into phishing and malicious email delivery attempts.

REC-ENDPOINT-01

Objective: Improve SOC transparency and investigative context for endpoint detections generated upstream.
Action: Correlate endpoint process and network telemetry with upstream alerts to reconstruct behavioral context where feasible.
Expected Outcome: Improved investigative clarity and reduced reliance on opaque verdicts.

REC-IDENTITY-01

Objective: Improve explainability and consistency of identity anomaly detection outcomes.
Action: Complement probabilistic detections with deterministic analytics and formalize triage guidance.
Expected Outcome: Increased analyst confidence and consistent handling of identity alerts.

9.3 Roadmap Phasing

Phase	Timeframe	Focus	Recommendations
Phase 1	0–3 months	Highest-risk detection gaps	REC-EMAIL-01
Phase 2	3–6 months	Visibility and ownership gaps	REC-ENDPOINT-01
Phase 3	6–12 months	Detection quality and consistency	REC-IDENTITY-01

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10. Non-Assertions

This report does not assert that existing security controls are ineffective or misconfigured. Findings reflect detection visibility and ownership boundaries at the SOC layer and do not represent a breach analysis or incident response readiness assessment.